

Leon E Etiobi

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MEng Systems Engineering graduate from the University of Warwick, experience in machine learning, computer vision and closed-loop control for autonomous systems. Project experience spans building a neural network from scratch in C++, fine-tuning object detection models for autonomous driving, and designing vehicle control systems in simulation.

TECHNICAL PROJECTS

Autonomous Vehicle Object Detection | Python, PyTorch, YOLOv10L [\[GitHub\]](#) 2024 – 2025

- Trained and benchmarked three sizes of the YOLOv10 vision model on real driving images (nulimages dataset) to detect pedestrians, vehicles and cyclists in low-light and bad-weather scenes.
- Diagnosed why the model missed objects using per-class error analysis and a confusion matrix, then rebalanced the three training loss weights to prioritise finding objects over perfecting box placement.
- Improved final accuracy to **71.4% mAP@50** and raised the share of real objects detected (recall 0.545→0.556) at a small cost in false alarms (precision 0.850→0.840): a deliberate safety trade-off.

Torque Vectoring Control for a Formula Student EV | MATLAB/Simulink, IPG CarMaker 2025 – 2026

- Developed a two-layer closed-loop controller that distributes torque between an electric race car's wheels to improve cornering, using Pacejka tyre modelling and load-transfer effects; worked in a four-person team and co-authored a 40-page group dissertation.
- Simulated performance across AWD, RWD, in-hub and in-board drivetrain configurations, targeting improved cornering stability and reduced trajectory error versus equal-torque control.

Neural Network from Scratch | C++, Eigen [\[GitHub\]](#) 2024

- Built a neural network entirely from scratch in C++ (no ML libraries), including custom forward/backward propagation, mini-batch gradient descent and parallelised training; it correctly reads **97.4% of handwritten digits (MNIST)**.

Reinforcement Learning Snake Agent | Python, PyTorch [\[GitHub\]](#) 2025

- Trained a deep Q-learning agent to play Snake for the Warwick AI competition, engineering a 15-channel spatial state encoding and reward shaping to reach a CI-verified average score of 26.3 across four difficulty tiers.

EDUCATION

University of Warwick 2022 – 2026

MEng Systems Engineering, 2:1 (Hons)

Brampton Manor Academy 2020 – 2022

A-Levels: Mathematics (A*), Further Mathematics (A), Physics (A)

EXPERIENCE

CPower Energy | Intern 2024

- Conducted supplier market research against technical specifications and reviewed tender documents to refine proposals; gained exposure to negotiation and pricing strategy.

SKILLS

Programming: Python, C++, MATLAB, Git

ML & Data Science: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, model training & evaluation, computer vision

Engineering & Simulation: Simulink, IPG CarMaker, closed-loop control design, system identification, vehicle dynamics, Arduino

VOLUNTEERING

Casework interviews with asylum seekers (RAMA Pamoja); administration and budgeting for community programmes (AfiUK).